ADVENTURES IN BIG MATH · 5TH GRADE MATH

MISSION 07 OF 08 · WORKSHEET PACK

Volume & Measure

Every graded set, every week — 4 weeks, 20 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Converting Units

Every conversion is a power of ten.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

1 m in cm	1 km in m	1 kg in g
2 m in cm	3 km in m	5 L in mL

Core GRADED · SHOW YOUR WORKING

1. 250 cm in m ....
2. 1,500 g in kg ....
3. 0.75 km in m ....
4. 3,200 mL in L ....
5. 45 mm in cm ....

Challenge NOT GRADED

C1. 2.5 kg in mg — type the number

C2. Add 1.2 m and 85 cm — answer in cm

NAME __________

DATE __________

SCORE ____ / 11

Counting Cubes

Volume is a count before it is a formula.

Warm-Up

6 ITEMS

A 2 by 2 by 2 cube — volume	A 3 by 1 by 1 box	A 2 by 3 by 1 box
A 4 by 2 by 1 box	A 3 by 3 by 1 box	A 1 by 1 by 5 box

Core

GRADED · SHOW YOUR WORKING

1. A 4 by 3 by 2 box ....
2. A 5 by 4 by 3 box ....
3. One layer of a 5 by 4 box holds ....
4. That box 3 layers high ....
5. A 6 by 5 by 2 box ....

Challenge

NOT GRADED

C1. A 10 by 8 by 4 box

C2. Volume 48, base 4 by 3 — the height

Volume Formulas

Length × width × height, and a missing edge you have to find.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

2 × 3 × 4	5 × 5 × 2	10 × 2 × 3
1 × 7 × 4	6 × 2 × 2	3 × 3 × 3

Core

GRADED · SHOW YOUR WORKING

1. Volume 120, base area 20 — the height ....
2. Volume 100, height 5 — the base area ....
3. A 12 by 5 by 4 tank ....
4. Volume 72, edges 6 and 3 — the third edge ....
5. A cube of edge 4 ....

Challenge

NOT GRADED

C1. Two boxes: 4×3×2 and 5×2×2 — total volume

....

C2. A cube of volume 125 — its edge

....

Line Plots

Fractional measurements, plotted and then reasoned about.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

Four items at 1/2 each — total	Eight items at 1/4 each — total	Two items at 3/4 each — total, as a/b
Three items at 1/3 each	Six items at 1/2 each	Longest of 1/4, 1/2, 3/8 — type it

Core

GRADED · SHOW YOUR WORKING

1. 1/2 + 1/4 + 1/4 — total
2. Range of 1/8 and 7/8 — type as a/b
3. Five items totalling 5/2 — the mean, as a/b
4. Difference between 7/8 and 3/8 — as a/b
5. Four measurements of 3/4 — total

Challenge

NOT GRADED

C1. Six pencils totalling 9/2 inches, shared equally — each, as a/b

C2. Two at 1/8, three at 1/4, one at 1/2 — total, as a/b

Box It

Enrichment. Same volume, different shapes.

NAME __________
DATE __________
SCORE ____ / 4

Warm-Up

2 ITEMS

A box of volume 24: 2 by 3 by ?

A box of volume 24: 1 by 4 by ?

Core

GRADED · SHOW YOUR WORKING

1. A box of volume 36: 3 by 3 by ?

2. A box of volume 60: 5 by 4 by ?

Challenge

NOT GRADED

C1. How many whole boxes of volume 8 fit in one of volume 96

C2. A cube with the same volume as a 2×4×8 box — its edge

C3. Double every edge of a 2×3×4 box. Volume multiplies by

TEACHER PAGE · ANSWER KEY

Week 1 Answers

Mission 07 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 · Converting Units

OUT OF 11

Warm-Up · 100 | 1000 | 1000 | 200 | 3000 | 5000

Core · 2.5 | 1.5 | 750 | 3.2 | 4.5

Challenge · 2500000 | 205

1.2 · Counting Cubes

OUT OF 11

Warm-Up · 8 | 3 | 6 | 8 | 9 | 5

Core · 24 | 60 | 20 | 60 | 60

Challenge · 320 | 4

1.3 · Volume Formulas

OUT OF 11

Warm-Up · 24 | 50 | 60 | 28 | 24 | 27

Core · 6 | 20 | 240 | 4 | 64

Challenge · 44 | 5

1.4 · Line Plots

OUT OF 11

Warm-Up · 2 | 2 | 3/2 | 1 | 3 | 1/2

Core · 1 | 3/4 | 1/2 | 1/2 | 3

Challenge · 3/4 | 3/2

Fri · Box It

OUT OF 4

Warm-Up · 4 | 6

Core · 4 | 3

Challenge · 12 | 4 | 8

Count the Cubes

Build it, count it, write it. The formula comes later.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

A 2 by 2 by 2 cube	A 3 by 1 by 1 box	A 2 by 3 by 1 box
A 4 by 2 by 1 box	A 3 by 3 by 1 box	A 1 by 1 by 5 box

Core GRADED · SHOW YOUR WORKING

1. A 4 by 3 by 2 box
2. A 5 by 4 by 3 box
3. One layer of a 5 by 4 box
4. That box 3 layers high
5. A 6 by 5 by 2 box

Challenge NOT GRADED

C1. A 10 by 8 by 4 box

C2. Volume 48, base 4 by 3 — the height

Layers

One layer, times the height. That is the whole idea.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up
6 ITEMS

A 3 by 4 base — cubes in one layer	Two such layers	A 5 by 2 base — one layer
Three such layers	A 6 by 1 base — one layer	Four such layers

Core
GRADED · SHOW YOUR WORKING

1. A 7 by 3 base, 4 high
2. A 8 by 5 base, 2 high
3. Base area 20, height 6
4. Base area 15, height 4
5. Volume 100, base area 20 — the height

Challenge
NOT GRADED

C1. Volume 288, base 12 by 4 — the height

C2. Base area 36, height 10

The Formula

Length × width × height, and where each one comes from.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

$2 \times 3 \times 4$	$5 \times 5 \times 2$	$10 \times 2 \times 3$
$1 \times 7 \times 4$	$6 \times 2 \times 2$	$3 \times 3 \times 3$

Core

GRADED · SHOW YOUR WORKING

1. A 12 by 5 by 4 tank
2. A cube of edge 4
3. A 9 by 3 by 2 box
4. A cube of edge 5
5. A 20 by 10 by 5 box

Challenge

NOT GRADED

C1. A cube of volume 125 — its edge

C2. A cube of volume 216 — its edge

Find a Missing Edge

Given volume and two edges, work out the third.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

Volume 12, edges 2 and 3 — the third	Volume 24, edges 2 and 3	Volume 30, edges 5 and 3
Volume 8, edges 2 and 2	Volume 20, edges 5 and 2	Volume 36, edges 6 and 3

Core GRADED · SHOW YOUR WORKING

1. Volume 72, edges 6 and 3 ....
2. Volume 120, edges 5 and 4 ....
3. Volume 240, edges 12 and 5 ....
4. Volume 100, height 5 — the base area ....
5. Volume 96, edges 8 and 4 ....

Challenge NOT GRADED

C1. Volume 360, edges 9 and 5

C2. Volume 1000 with equal edges — one edge

Cube Count

Read the drawing, call the volume before your opponent does.

NAME __________

DATE __________

SCORE ____ / 4

Warm-Up

2 ITEMS

A 2 by 2 by 3 box

A 3 by 3 by 2 box

Core

GRADED · SHOW YOUR WORKING

1. A 4 by 4 by 4 cube

2. A 5 by 3 by 4 box

Challenge

NOT GRADED

C1. Two boxes: 4×3×2 and 5×2×2 — total

C2. Double every edge of a 2×3×4 box — volume multiplies by

C3. A 6×6×6 cube

TEACHER PAGE · ANSWER KEY

Week 2 Answers

Mission 07 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 · Count the Cubes

OUT OF 11

Warm-Up · 8 | 3 | 6 | 8 | 9 | 5

Core · 24 | 60 | 20 | 60 | 60

Challenge · 320 | 4

2.2 · Layers

OUT OF 11

Warm-Up · 12 | 24 | 10 | 30 | 6 | 24

Core · 84 | 80 | 120 | 60 | 5

Challenge · 6 | 360

2.3 · The Formula

OUT OF 11

Warm-Up · 24 | 50 | 60 | 28 | 24 | 27

Core · 240 | 64 | 54 | 125 | 1000

Challenge · 5 | 6

2.4 · Find a Missing Edge

OUT OF 11

Warm-Up · 2 | 4 | 2 | 2 | 2 | 2

Core · 4 | 6 | 4 | 20 | 3

Challenge · 8 | 10

Fri · Cube Count

OUT OF 4

Warm-Up · 12 | 18

Core · 64 | 60

Challenge · 44 | 8 | 216

Split the Solid

Draw the dividing line first, then measure each part.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

A 2 by 2 by 2 plus a 2 by 2 by 1	A 3 by 2 by 1 plus a 3 by 2 by 1	A 4 by 2 by 2 plus a 2 by 2 by 2
A 5 by 1 by 1 plus a 3 by 1 by 1	Parts in an L-shaped solid	A 2 by 2 by 5 box

Core GRADED · SHOW YOUR WORKING

1. A 6×4×2 plus a 3×4×2
2. A 10×5×2 minus a 4×5×2
3. An L of 8×3×2 and 4×3×2
4. A 5×5Õ4 minus a 2×2Õ4
5. A T of 6×2×2 and 2×4×2

Challenge NOT GRADED

C1. A 10×10×5 with a 4×4×5 hole

C2. A 12×6×3 plus a 6×6×3

Add the Volumes

Two boxes, one total. Check each part before adding.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

12 + 8	24 + 16	A 2×2×2 and a 3×2×2 — total
A 4×2×1 and a 4×2×1	30 + 45	A 5×2×2 and a 5×2×2

Core

GRADED · SHOW YOUR WORKING

1. A 6×5×2 and a 4×3×2
2. A 8×4×3 and a 2×2×3
3. A 10×3×2 and a 5×3×2
4. Three 4×3×2 boxes
5. A 7×4×2 and a 3×4×2

Challenge

NOT GRADED

C1. A 12×8×4 and a 6×4×4

C2. Five 6×5×2 boxes

Line Plots

Plot measurements to the nearest eighth, then reason about them.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

Four items at 1/2 — total	Eight items at 1/4 — total	Three items at 1/3 — total
Six items at 1/2 — total	Two items at 3/4 — total	Longest of 1/4, 1/2, 3/8

Core

GRADED · SHOW YOUR WORKING

1. 1/2 + 1/4 + 1/4
2. Range of 1/8 and 7/8
3. Five items totalling 5/2 — the mean
4. Difference between 7/8 and 3/8
5. Four measurements of 3/4 — total

Challenge

NOT GRADED

C1. Six pencils totalling 9/2 inches shared equally — each

C2. Two at 1/8, three at 1/4, one at 1/2 — total

Read the Plot

Total, difference, and redistribution.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up

6 ITEMS

Five values of 1/2 — total	Their mean	Four values of 1/4 — total
Their mean	Range of 1/4 and 3/4	Range of 1/8 and 5/8

Core

GRADED · SHOW YOUR WORKING

1. Values 1/4, 1/2, 3/4 — total
2. Their mean
3. Values 1/8, 3/8, 1/2 — total
4. Four values totalling 3 — the mean
5. Range of 1/8 and 1

Challenge

NOT GRADED

C1. Eight values totalling 5 — the mean

C2. Redistribute 9/2 across 6 equally — each

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

NAME __________
DATE __________
SCORE ____ / 7

Warm-Up 2 ITEMS

1 m in cm

A 2 by 3 by 4 box

Core GRADED · SHOW YOUR WORKING

1. 250 cm in m
2. 1500 g in kg
3. A 5 by 4 by 3 box
4. Volume 120, base area 20 — the height
5. Four measurements of 3/4 — total

Challenge NOT GRADED

C1. A 10×10×5 with a 4×4×5 hole

TEACHER PAGE · ANSWER KEY

Week 3 Answers

Mission 07 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 · Split the Solid

OUT OF 11

Warm-Up · 12 | 12 | 24 | 8 | 2 | 20

Core · 72 | 60 | 72 | 84 | 40

Challenge · 420 | 324

3.2 · Add the Volumes

OUT OF 11

Warm-Up · 20 | 40 | 20 | 16 | 75 | 40

Core · 84 | 108 | 90 | 72 | 80

Challenge · 480 | 300

3.3 · Line Plots

OUT OF 11

Warm-Up · 2 | 2 | 1 | 3 | 3/2 | 1/2

Core · 1 | 3/4 | 1/2 | 1/2 | 3

Challenge · 3/4 | 3/2

3.4 · Read the Plot

OUT OF 11

Warm-Up · 5/2 | 1/2 | 1 | 1/4 | 1/2 | 1/2

Core · 3/2 | 1/2 | 1 | 3/4 | 7/8

Challenge · 5/8 | 3/4

Fri · Mid-Unit Quiz

OUT OF 7

Warm-Up · 100 | 24

Core · 2.5 | 1.5 | 60 | 6 | 3

Challenge · 420

Design Three Boxes

Same volume, different shapes. All three must check out.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up

6 ITEMS

A box of volume 24: 2 by 3 by ?	Volume 24: 1 by 4 by ?	Volume 24: 2 by 2 by ?
Volume 12: 2 by 3 by ?	Volume 36: 3 by 3 by ?	Volume 60: 5 by 4 by ?

Core

GRADED · SHOW YOUR WORKING

1. Volume 48: 4 by 4 by ?
2. Volume 72: 6 by 3 by ?
3. Volume 96: 8 by 4 by ?
4. Volume 120: 6 by 5 by ?
5. Volume 64 with equal edges — one edge

Challenge

NOT GRADED

C1. A cube with the same volume as a 2×4×8 box — its edge

C2. How many whole boxes of volume 8 fit in one of volume 96

Argue for One

Which shape is best, and best for what?

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up
6 ITEMS

A 2×2×6 box — volume	A 2×3×4 box — volume	A 1×4×6 box — volume
Are those volumes equal — yes or no	A 24 cm³ box on a 12 cm² base — its height	A cube of edge 3 — volume

Core
GRADED · SHOW YOUR WORKING

1. Surface area of a 2×2×6 box ....
2. Surface area of a 2×3×4 box ....
3. Which uses less card — type 2x2x6 or 2x3x4 ....
4. Surface area of a cube of edge 3 ....
5. Surface area of a 1×4×6 box ....

Challenge
NOT GRADED

C1. For a fixed volume, the shape using least material is nearest a — type cube or slab

C2. Surface area of a cube of edge 4

Conversions and Volume

Both strands together, which is how they arrive in real problems.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

1 m in cm	1 kg in g	2 m in cm
3 L in mL	A 2×2×2 box	500 g in kg

Core GRADED · SHOW YOUR WORKING

1. 250 cm in m
2. 1500 g in kg
3. A 100 by 50 by 20 cm box — volume in cubic cm
4. 3200 mL in L
5. A 2 by 1 by 0.5 m tank — volume in cubic m

Challenge NOT GRADED

C1. 1000 cubic cm in litres

C2. A 2 by 1 by 0.5 m tank in litres

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up
6 ITEMS

1 km in m	A 3 by 3 by 3 cube	1 L in mL
A 4 by 2 by 1 box	2 kg in g	Four items at 1/2 — total

Core
GRADED · SHOW YOUR WORKING

1. 45 mm in cm
2. A 5 by 4 by 3 box
3. Volume 72, edges 6 and 3 — the third
4. 0.75 km in m
5. Range of 1/8 and 7/8

Challenge
NOT GRADED

C1. A cube of volume 216 — its edge

C2. Add 1.2 m and 85 cm — answer in cm

Mission 07 Test

Twelve items plus the Big Question, answered out loud.

NAME __________
DATE __________
SCORE ____ / 10

Warm-Up
2 ITEMS

1 m in cm

A 2 by 3 by 4 box

Core
GRADED · SHOW YOUR WORKING

1. 250 cm in m
2. 3200 mL in L
3. A 5 by 4 by 3 box
4. A cube of edge 4
5. Volume 120, base area 20 — the height
6. Volume 96, edges 8 and 4 — the third
7. A 6×5×2 and a 4×3×2 — total
8. Five items totalling 5/2 — the mean

Challenge
NOT GRADED

C1. A cube of volume 125 — its edge

C2. Double every edge of a 2×3×4 box — volume multiplies by

TEACHER PAGE · ANSWER KEY

Week 4 Answers

Mission 07 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 · Design Three Boxes

OUT OF 11

Warm-Up · 4 | 6 | 6 | 2 | 4 | 3

Core · 3 | 4 | 3 | 4 | 4

Challenge · 4 | 12

4.2 · Argue for One

OUT OF 11

Warm-Up · 24 | 24 | 24 | yes | 2 | 27

Core · 56 | 52 | 2x3x4 | 54 | 68

Challenge · cube | 96

4.3 · Conversions and Volume

OUT OF 11

Warm-Up · 100 | 1000 | 200 | 3000 | 8 | 0.5

Core · 2.5 | 1.5 | 100000 | 3.2 | 1

Challenge · 1 | 1000

Thu · Error Journal Sweep

OUT OF 11

Warm-Up · 1000 | 27 | 1000 | 8 | 2000 | 2

Core · 4.5 | 60 | 4 | 750 | 3/4

Challenge · 6 | 205

Fri · Mission 07 Test

OUT OF 10

Warm-Up · 100 | 24

Core · 2.5 | 3.2 | 60 | 64 | 6 | 3 | 84 | 1/2

Challenge · 5 | 8